444 Box 12.5: Davies (1975), Unruh (1976), Unruh and Wald (1982, 1984).

446 [a highly abstract proof . . . in 1977.] Gibbons and Hawking (1977).

446 [The total lifetime, . . . Don Page] Page (1976).

447 [Detailed calculations by Hawking, . . . to produce tiny holes.] E.g., Hawking (1971a); Novikov, Polnarev, Starobinsky, and Zel'dovich (1979).

447 [The absence of excess gamma rays . . . soft equation of state.] Page and Hawking (1975); Novikov, Polnarev, Starobinsky, and Zel'dovich (1979).

CHAPTER 13

449 General comment about Chapter 13: The historical aspects of this chapter are based on (i) my own personal experience (though as an observer rather than a participant), (ii) my interviews with participants (INT-Belinsky, INT-DeWitt, INT-Geroch, INT-Khalatnikov, INT-Lifshitz, INT-MacCallum, INT-Misner, INT-Penrose, INT-Sciama, INT-Wheeler), and (iii) my reading of scientific papers the participants wrote.

499 [John Archibald Wheeler taught . . . outside the horizon.] Harrison, Wakano, and Wheeler (1958); Wheeler (1960).

450 [Wheeler retained his conviction . . . pursuing.] Wheeler (1964a,b); Harrison, Thorne, Wakano, and Wheeler (1965).

450 [J. Robert Oppenheimer and Hartland Snyder,] Oppenheimer and Snyder (1939).

450 [Perhaps Oppenheimer's unwillingness to speculate,] See the last several pages of Chapter 5.

451 [The singularity predicted by the Oppenheimer–Snyder calculations] The singularity as described here is that in the vacuum outside the imploding star, and since the vacuum region is described by the Schwarzschild solution of Einstein's equations, this singularity is often referred to as the *singularity of the Schwarischild geometry.* It is analyzed quantitatively, e.g., in Chapter 32 of MTW.

451 Figure 13.1: Ibid.

453 [One group, . . . general relativity fails] Wheeler (1960, 1964a,b); Harrison, Thorne, Wakano, and Wheeler (1965).

453 [A second group, . . . Khalatnikov and Evgeny Michailovich Lifshitz . . . could not be trusted.] This viewpoint and the calculations that led Khalatnikov and Lifshitz to it were published in Lifshitz and Khalatnikov (1960, 1963) and in Landau and Lifshitz (1962).

454 [Khalatnikov and Lifshitz . . . *small perturbations*.] Ibid.

456 [*The Classical Theory of Fields*.] Landau and Lifshitz (1962).

457 Figure 13.4: It was obvious in the early 1960s to students in Wheeler's group, where the Graves–Brill (1960) research had been done, that there must exist a solution to Einstein's equations of the sort depicted here. However, I gather from a discussion with Penrose that researchers in most other groups did not become aware of it until the late 1960s. It was difficult to construct such solutions explicitly, and we in Wheeler's group did not try, and did not publish anything on the issue. The first publication of the idea and the first attempt at an explicit solution, so far as I know, were by Novikov (1966).

458 [Hans Reissner and Gunnar Nordström . . . Dieter Brill and John Graves,] Graves and Brill (1960) and references therein.

459 [Roger Penrose grew up in a British . . .] This biographical discussion of Penrose comes largely from INT-Penrose and INT-Sciama.